

Processando config 1, seed 4, teste: Rabbit
XX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

smultin_MultinomialBitsOver test:

N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = $1 / 32768$
EColl = $n^2 / (2k)$ = 128
Hashing = TRUE

Collision test
CollisionOver: density = $n / k = 1 / 32768.00$
Expected number of collisions = $\mu = 128.00$

Results of CollisionOver test:

POISSON approximation :
Expected number of collisions = $N \cdot \mu$: 256.00
Observed number of collisions : 312
p-value of test : $3.8e-4$ *****

Total number of cells containing j balls

j = 0	:	549739036984
j = 1	:	16776592
j = 2	:	312
j = 3	:	0
j = 4	:	0
j = 5	:	0

CPU time used : 00:00:04.57

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 18.00
p-value of test : 0.12

CPU time used : 00:00:00.06

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 12.00
p-value of test : 1.9e-6 *****

CPU time used : 00:00:00.06

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.63
p-value of test : 0.74

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 11.35
p-value of test : 0.12

Normal statistic for number of jumps : -52.71
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.06

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -1.15
p-value of test : 0.88

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -1.34
p-value of test : 0.91

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.032
p-value of test : 0.34

Kolmogorov-Smirnov- statistic = D- : 0.022
p-value of test : 0.59

Anderson-Darling statistic = A2 : 0.42
p-value of test : 0.83

CPU time used : 00:00:00.11

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 6.70
p-value of test : 0.08

Global longest run of 1 : 22.00
p-value of test : 0.63

CPU time used : 00:00:00.07

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 123.26
p-value of test : 0.02

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 20.73
p-value of test : 0.54

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.35
p-value of test : 0.36

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.80
p-value of test : 0.79

CPU time used : 00:00:00.02

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.89
p-value of test : 0.19

CPU time used : 00:00:00.02

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 199.50
p-value of test : 0.12

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 290.20
p-value of test : 0.72

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 2380.96
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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-----
Normal statistic          :   -0.97
p-value of test          :    0.83
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 269499904 bits have been read.
```

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*****
```

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HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :   -1.61
p-value of test          :    0.95
-----
```

```
CPU time used            :   00:00:00.02
```

```
Generator state:
 286277088 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin
```

```
sstring_Run test:
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```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355447
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :   34.61
p-value of test                   :    0.53
-----
```

```
-----
Total number of bits:  13417920
```

```
Normal statistic for number of bits :   -0.74
p-value of test                   :    0.77
```

CPU time used : 00:00:00.06

Generator state:
299695008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 92.88
p-value of test : eps *****

CPU time used : 00:00:00.06

Generator state:
316472224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333163424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42	
ChiSquare statistic	:	963.70	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	42	
ChiSquare statistic	:	457.19	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	64	
ChiSquare statistic	:	195.96	
p-value of test	:	2.7e-15	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	36	
ChiSquare statistic	:	41.45	
p-value of test	:	0.24	

Test on the values of the Statistic C

Number of degrees of freedom	:	22	
ChiSquare statistic	:	30.11	
p-value of test	:	0.12	

CPU time used : 00:00:00.04

Generator state:
349940640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94	
ChiSquare statistic	:	469.07	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	94	
ChiSquare statistic	:	494.01	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	512	
ChiSquare statistic	:	638.26	
p-value of test	:	1.2e-4	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	90	
ChiSquare statistic	:	94.43	
p-value of test	:	0.35	

Test on the values of the Statistic C

Number of degrees of freedom	:	50	
ChiSquare statistic	:	47.52	
p-value of test	:	0.57	

CPU time used : 00:00:00.04

Generator state:

366717856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed4.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	151.53
p-value of test	:	0.10

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	155.29
p-value of test	:	0.07

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	176.47
p-value of test	:	0.19

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	146.39
p-value of test	:	0.15

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	106.59
p-value of test	:	0.10

CPU time used : 00:00:00.03

Generator state:

383494656 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.07
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
1	MultinomialBitsOver	3.8e-4
3	ClosePairsBitMatch, t = 4	1.9e-6
5	LinearComp	1 - eps1
17	HammingIndep, L = 64	eps
21	MatrixRank, 32 x 32	eps
22	MatrixRank, 320 x 320	eps
24	RandomWalk1 H	eps
24	RandomWalk1 M	eps
24	RandomWalk1 J	2.7e-15
25	RandomWalk1 H (L = 1024)	eps
25	RandomWalk1 M (L = 1024)	eps
25	RandomWalk1 J (L = 1024)	1.2e-4

All other tests were passed